Toh Wei Xuan

Computer Science (Real-Time Interactive Simulation) student · Former Air Force Engineer

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PROFILE

Computer Science student specializing in Real-Time Interactive Simulation, building software in C++ and C with a focus on system architecture, state management, tooling and debugging. Designed the Entity Component System, event system, scene serialization and editor undo/redo for a team-built C++ engine. Former Air Force Engineer with four years in a high-reliability engineering environment, bringing systematic troubleshooting, quality control and team leadership.

EDUCATION

Singapore Institute of Technology

September 2024 – Present

Computer Science in Real-Time Interactive Simulation

PROJECTS

Soot & Scoot — Engine Champion

Sep 2025 – Apr 2026

GAM 200/250 · Team of 8 · C++, OpenGL, ECS, ImGui, RTTR, Lua (sol2), FMOD, CMake

- Designed and refactored the engine's core Entity Component System (ECS), engine loop, window manager and event manager in C++.
- Built enemy AI using A* pathfinding on the tilemap with context steering for local obstacle avoidance, plus line-of-sight detection and patrol, look-around, chase and return states.
- Implemented scene serialization, a prefab system with hot-reload, a tilemap editor and generic RTTR-based undo/redo in the ImGui level editor.
- Developed a particle system (continuous, burst and trail emitters) and an MPEG video player with audio synced through FMOD for cutscenes.
- Integrated Lua scripting via sol2, and built the game's scene flow with controller support, fullscreen toggle and pause on focus loss.

Dwarf Mayhem: Into the Depths — Gameplay Programmer

Jan – Mar 2025

CSD1451 · Team of 5 · C++, Alpha Engine, cellular automata

- Implemented player mining on a pixel-based cellular simulation, with line-traced cell selection, a highlighting smart cursor and an upgradable mining area.
- Built the inventory system, inventory scene and shop scene for selling ores and buying upgrades (mining size, jetpack, heat armor, full heal).
- Designed depth-scaled heat damage absorbed by heat armor, with an armor bar and a depth indicator in the UI.

Portal Mayhem — Weapons Programmer

Sep – Nov 2024

CSD1401 · Team of 5 · C, CProcessing

- Led development of the weapon system in C, covering the portal gun, pistol, grenade launcher and disk launcher.
- Implemented portal shooting and player teleportation that carries angle and velocity through the exit portal, with a teleport cooldown.
- Built projectile behaviours including gravity-affected bouncing grenades, disks that split in two after passing through a portal, and area-damage explosions.

EXPERIENCE

Air Force Engineer

August 2019 – August 2023

Republic of Singapore Air Force (RSAF) · Singapore

- Maintained the serviceability of fighter aircraft within a high-reliability engineering environment.
- Applied systematic troubleshooting and problem-solving to aircraft maintenance activities.
- Implemented new procedures and technologies that improved efficiency and streamlined operations.
- Led a team to achieve expected maintenance goals while reducing human-factor errors.
- Applied quality control, technical procedures and effective team communication to maintain operational standards.

TECHNICAL SKILLS

- **Programming:** C++, C, Lua, OpenGL
- **Software Engineering:** Object-Oriented Programming, ECS Architecture, Event-Driven Systems, Debugging
- **Game Systems:** Real-Time Simulation, Gameplay Programming, Collision Detection, A* Pathfinding, AI / Steering Behaviour, Particle Systems, Serialization & Reflection, Scene Management, UI Integration
- **Tools:** Git, CMake, Visual Studio, ImGui, FMOD
- **Professional:** Team Collaboration, Problem Solving, Technical Documentation, Technical Reporting
- **Languages:** English (Fluent), Chinese (Native)